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PUSL3190 Computing Project

Abstract

An Automated Vehicle Attendance and
Movement Analytics System for Industrial
Facilities Using Computer Vision

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Abstract

Industrial facilities that rely on contractor vehicle fleets face a persistent challenge in accurately recording vehicle attendance. Manual logbooks are error-prone and falsifiable, while radio-frequency identification gate systems experience read failures of 15 to 20 percent under Sri Lankan port field conditions, producing records that are incomplete and indefensible in payroll disputes. This project designed, implemented, and evaluated an automated vehicle attendance and analytics system that uses computer vision to identify vehicles by their licence plates at facility gates, with no hardware required on individual vehicles. A custom-trained two-stage image recognition pipeline was developed for Sri Lankan alphanumeric plates, incorporating a domain-specific weighted error-correction algorithm to resolve visually ambiguous characters. A shift-aware attendance engine records arrival and departure events, computes time on-site, flags late arrivals and early departures, and handles unregistered vehicles through a structured operator alert workflow. An analytics dashboard generates daily, weekly, and monthly attendance reports with payroll-compatible exports, and all records are protected by a cryptographic hash chain with photographic plate evidence to ensure tamper-evident auditability. Evaluation was conducted against 150 physical test plates and 150 scripted attendance scenarios on a controlled two-gate testbed. Plate recognition accuracy reached 91.3%, compared to 71.0% produced by a widely used general-purpose text recognition tool under identical conditions - a 20 percentage-point improvement. Attendance event accuracy was 97.3%, a 12 to 18 percentage-point improvement over the existing radio-frequency gate system. Gate event processing was completed in under 300 milliseconds at the 95th percentile, well within the 500-millisecond operational target. The per-gate hardware cost of approximately 85,000 Sri Lankan Rupees is 40 times lower than commercially available alternatives, demonstrating that accurate, auditable vehicle attendance management is achievable at small enterprise scale.